Chapter 2 · Semester 1

KINEMATICS

Notes by Cikgu Suhaili

21 worked examples 3 simulations Exercises with answers

2.0 Introduction

Kinematics

is the study of **motion without regard to mass or force**. (From the Greek word for *motion*.)

In this chapter we study motion along

- **a straight line** (one-dimensional motion): linear motion, including **free fall**
- **a curved path** (two-dimensional motion): **projectile motion**

2.1 Linear motion

Distance and displacement

Distance, s

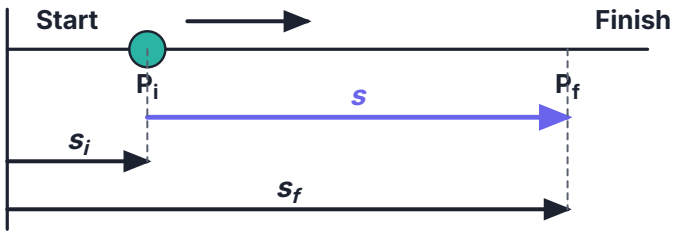
is the **length of the path** between two specified moments in time. It is a **scalar**.

Displacement, $\vec{s}$

is the **difference between the final position and the initial position** of a moving object. It is a **vector**.

$$s = s_f - s_i$$

where s_f = final position and s_i = initial position.



Speed and velocity

Speed, v

is the **rate of change of distance** (change of distance per unit time). Scalar.

$$v = \frac{\mathrm{d}s}{\mathrm{d}t}$$

Velocity, $\vec{v}$

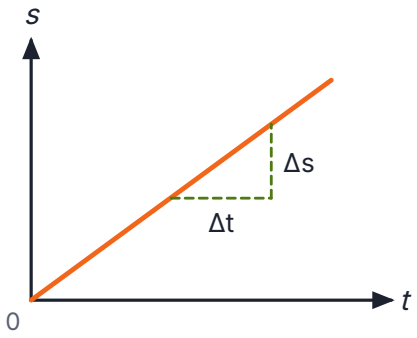
is the **rate of change of displacement** (change of displacement per unit time). Vector.

$$\vec{v} = \frac{\mathrm{d}s}{\mathrm{d}t}$$

Average speed, $\langle v \rangle$

is the **distance travelled divided by the elapsed time**.

$$\langle v \rangle = \frac{\Delta s}{\Delta t}$$



Average velocity, $\langle \vec{v} \rangle$

is the **displacement divided by the elapsed time**, in the same direction as the displacement.

$$\langle \vec{v} \rangle = \frac{\Delta s}{\Delta t} = \frac{s_\mathrm{f} - s_\mathrm{i}}{t_\mathrm{f} - t_\mathrm{i}}$$

Average velocity = gradient of the displacement–time graph:

$$\langle \vec{v} \rangle = \frac{\Delta s}{\Delta t}$$

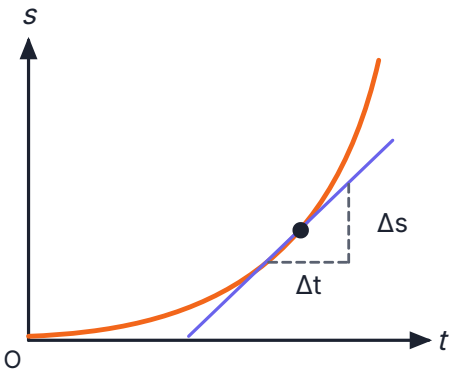
A **straight-line** s–t graph means the object moves with **uniform velocity**, so the velocity at every instant equals the average velocity.

Instantaneous velocity

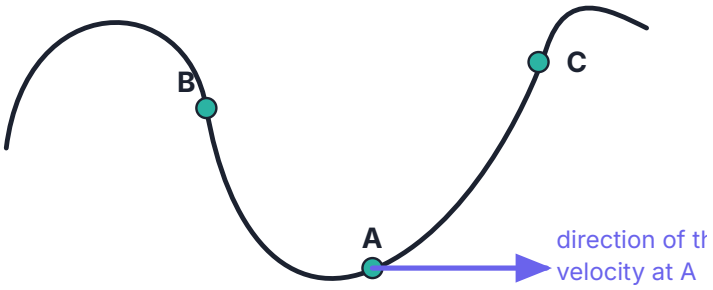
is the limit of the average velocity as the time interval approaches zero; it equals the instantaneous rate of change of displacement with time.

$$\vec{v} = \lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t} = \frac{\mathrm{d}s}{\mathrm{d}t}$$

= **gradient of the tangent** to the s–t graph at that instant.



- Instantaneous velocity is a **vector**.
- Its **direction** at any instant is ***along the tangent*** to the path at that point (e.g. point A).
- Its **magnitude** at any instant is equal to the ***speed*** at that point.



Uniform velocity

Velocity is uniform (**constant**) if (a) its magnitude is constant and (b) its direction is unchanged. The average and instantaneous velocities are then equal.

Acceleration

Acceleration, $\vec{a}$

is the **rate of change of velocity**.

$$\vec{a} = \frac{\mathrm{d}v}{\mathrm{d}t}$$

- **Acceleration** is the rate of ***increase*** of velocity: acceleration and velocity point in the same direction (positive value).
- **Deceleration** is the rate of ***decrease*** of velocity: acceleration and velocity point in opposite directions (negative value).

Average acceleration, $\langle\vec{a}\rangle$

is the **change in velocity divided by the time taken**. It shows how the speed and its direction change with time.

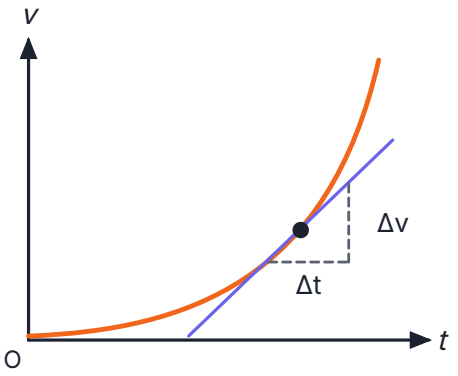
$$\langle\vec{a}\rangle = \frac{\Delta v}{\Delta t} = \frac{v_\mathrm{f} - v_\mathrm{i}}{t_\mathrm{f} - t_\mathrm{i}}$$

Instantaneous acceleration

is the limit of the average acceleration as the time interval approaches zero; it equals the instantaneous rate of change of velocity with time.

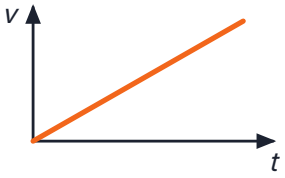
$$\vec{a} = \lim_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = \frac{dv}{dt}$$

= **gradient of the tangent** to the v–t graph.



Uniform acceleration

Acceleration is uniform (**constant**) if (a) its magnitude is constant and (b) its direction is unchanged. The average and instantaneous accelerations are then equal.



Velocity increases uniformly: gradient constant, so a is uniform.

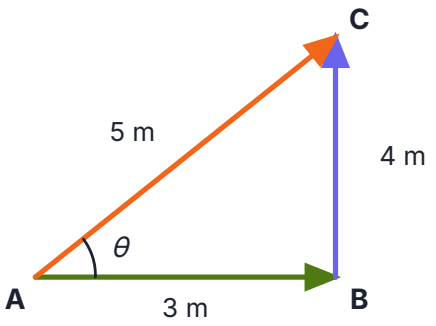


Constant velocity: gradient = 0, so $a = 0$.

Worked examples: distance, displacement, speed and velocity

Example 2.1.1

An object moves 3 m from A to B, then 4 m from B to C (at right angles). Find its distance and displacement.



Solution

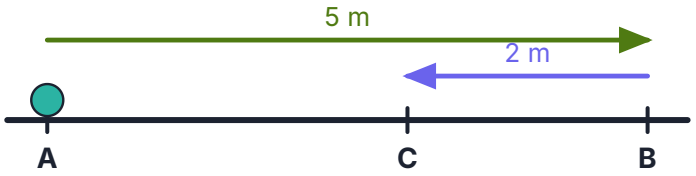
Distance = AB + BC = 3 + 4 = **7 m**

Displacement: magnitude AC = $\sqrt{3^2 + 4^2}$ = **5 m**

Direction: θ = $\cos^{-1}\frac{3}{5}$ = **53.1°** from AB

Example 2.1.2

An object moves from A to B (5 m) and then from B back to C (2 m). What is its distance? What is its displacement?



Solution

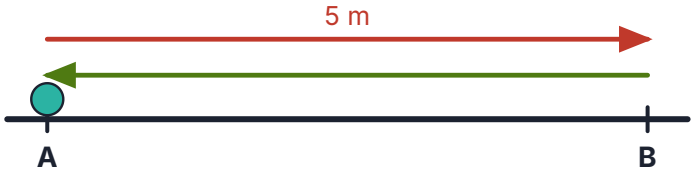
Distance = AB + BC = 5 + 2 = **7 m**

Displacement: AC = (+5) + (−2) = **+3 m**

Direction: **3 m to the right of A**

Example 2.1.3

An object moves from A to B (5 m) and then from B back to A. What are the distance and the displacement?



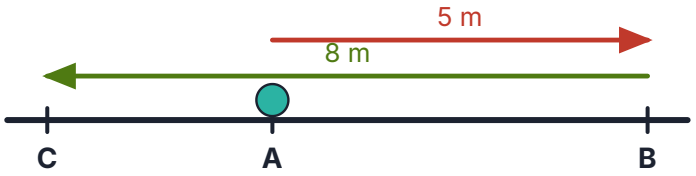
Solution

Distance = 5 + 5 = **10 m**

Displacement = (+5) + (−5) = **0 m**

Example 2.1.4

An object moves from A to B (5 m) and then from B to C (8 m). What is its distance? What is its displacement?



Solution

Distance = 5 + 8 = **13 m**

Displacement: AC = (+5) + (−8) = **−3 m**

Direction: **3 m to the left of A**

Speed, v

Scalar. Total distance moved, s, divided by the time, t.

$$v = \frac{s}{t}$$

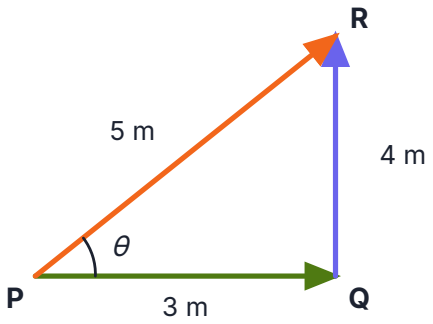
Velocity, $\vec{v}$

Vector. Displacement, s, divided by the time, t.

$$\vec{v} = \frac{s}{t} \text{ or } \vec{v} = \lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t} = \frac{\mathrm{d}s}{\mathrm{d}t}$$

Example 2.1.5

An object moves from P to Q (3 m) in 3 s and from Q to R (4 m, at right angles) in 2 s. What are its speed and velocity?



Solution

Distance = PQ + QR = 3 + 4 = 7 m; time = 3 + 2 = 5 s

Speed = $\frac{7}{5}$ = **1.4 m s^{-1}**

Displacement = PR = 5 m; time = 5 s

Velocity: magnitude = $\frac{5}{5}$ = **1 m s^{-1}**

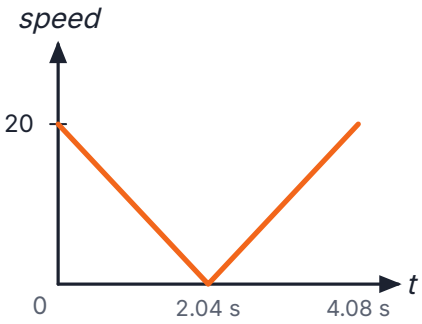
Direction: tan θ = $\frac{4}{3}$, θ = **53.1°** from PQ

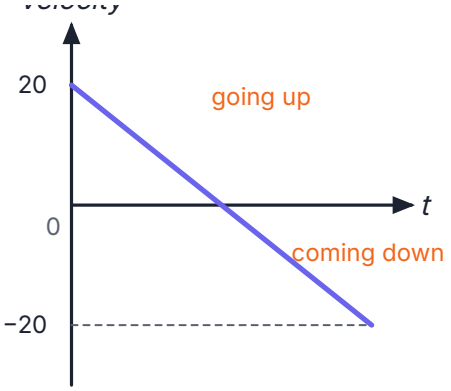
Example 2.1.6

A stone is thrown vertically upward with velocity u = 20 m s^{-1} and returns to its original position. Sketch the graphs of speed against time and velocity against time.

Solution

Going up, the speed falls from 20 m s^{-1} to 0 at the top; coming down it rises back to 20 m s^{-1}. The velocity changes uniformly from +20 m s^{-1} to −20 m s^{-1} (negative = downward); the gradient is $-g$ throughout.

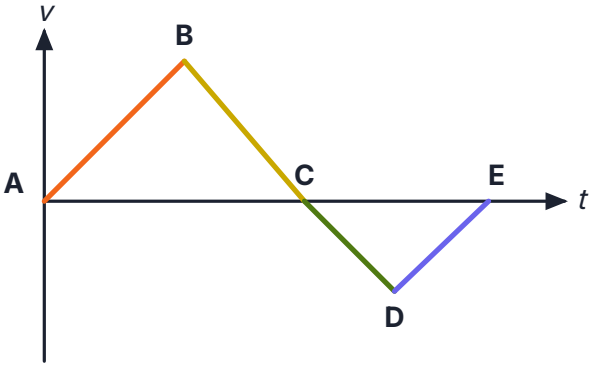




Acceleration and deceleration from a graph

Example 2.1.7

Based on the v–t graph below, describe the motion of the particle.

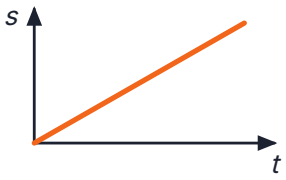
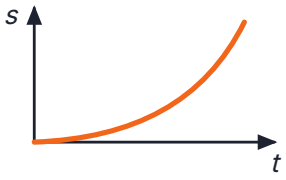
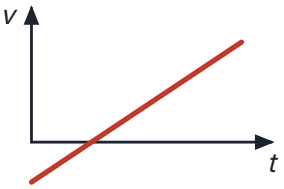
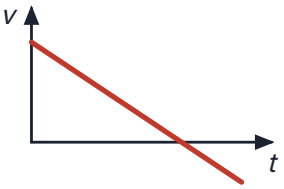
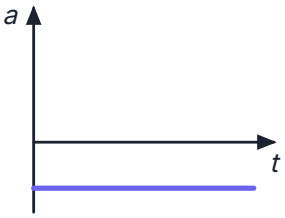


Solution

- **AB**: v increases ⇒ **acceleration** (constant)
- **BC**: v decreases ⇒ **deceleration** (constant)
- **CD**: v increases in the opposite direction ($-v$) ⇒ **acceleration**
- **DE**: v decreases, still in the opposite direction ($-v$) ⇒ **deceleration**

Graphs for linear motion

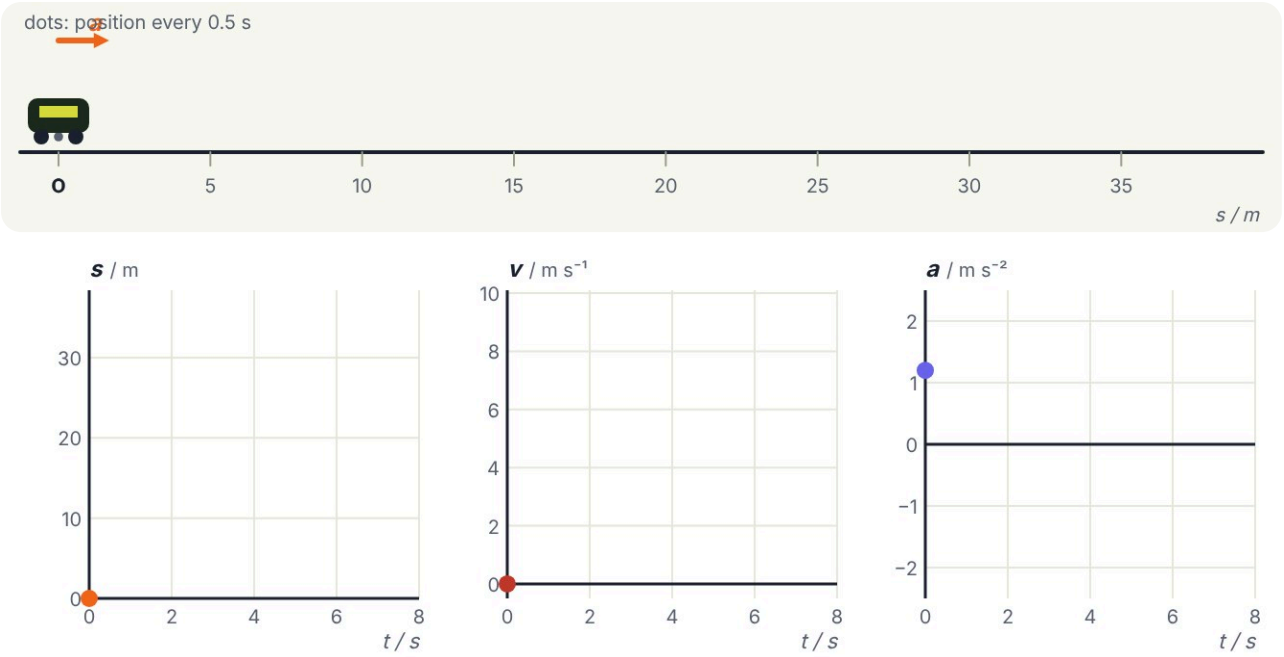
Gradient of s–t = velocity. Gradient of v–t = acceleration. Area under v–t = displacement. Area under a–t = change in velocity.

	s–t graph	v–t graph	a–t graph
Uniform velocity	 gradient = velocity (constant)	 gradient = 0, area = displacement	 a = 0
Uniform acceleration	 gradient increases: velocity increases	 velocity increases uniformly	 a positive, area = change in velocity
Uniform deceleration	 gradient decreases: velocity decreases	 velocity decreases uniformly	 a negative, area = change in velocity

Simulation

Motion graphs: watch s–t, v–t and a–t being drawn

Pick a motion, or set the initial velocity and acceleration yourself, then press Play. Turn on "Area under v–t" to see that the shaded area equals the displacement.

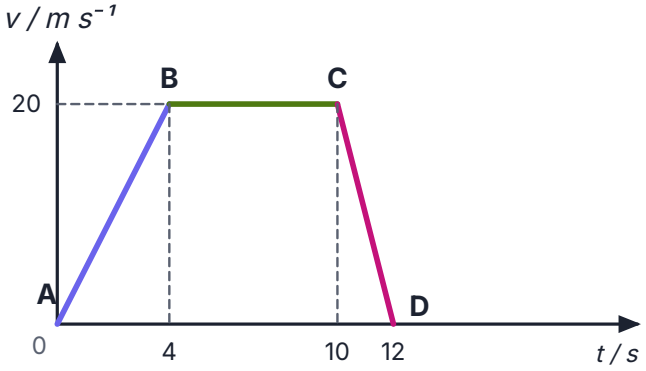


t	displacement s	velocity v	acceleration a
0.0 s	**0.0 m**	**0.0 m/s**	**1.2 m/s²**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 2 to use the simulation.

Example 2.1.8

From the v–t graph, describe the motion of the object and determine (a) the acceleration, (b) the deceleration, (c) the total distance moved.



Solution

- At A: $t = 0$, $v = 0$
- **AB**: v increases uniformly, constant acceleration
- **BC**: constant v, $a = 0$
- **CD**: v decreases uniformly, constant deceleration
- At D: $v = 0$

(a) Acceleration, $a = \dfrac{20}{4} = \mathbf{5\ m\,s^{-2}}$

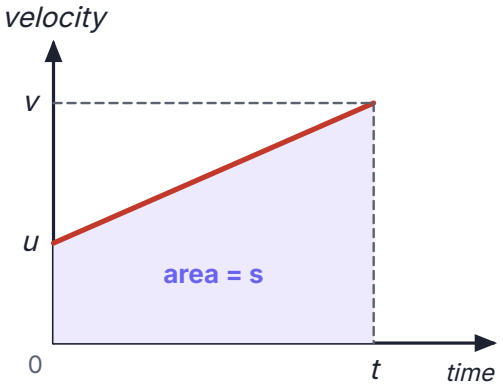
(b) Deceleration, $a = \dfrac{0 - 20}{12 - 10} = \mathbf{-10\ m\,s^{-2}}$

(c) Total distance = area under graph = $\dfrac{1}{2}[(10 - 4) + 12] \times 20 = \mathbf{180\ m}$

2.1 Equations of motion (uniform acceleration)

From the v–t graph:

- v = final velocity
- u = initial velocity
- a = acceleration
- t = time
- s = area under the graph = displacement



From $a = \frac{dv}{dt} = \frac{v-u}{t}$, rearranging: (i) $\boldsymbol{v = u + at}$

Area under the graph (trapezium): (ii) $\boldsymbol{s = \frac{1}{2}(u+v)t}$

Substitute (i) into (ii): $s = \frac{1}{2}[u + (u + at)]t$, so (iii) $\boldsymbol{s = ut + \frac{1}{2}at^2}$

From (i), $t = \frac{v-u}{a}$. Substitute into (iii): $s = u(\frac{v-u}{a}) + \frac{1}{2}a(\frac{v-u}{a})^2$

$2as = 2uv - 2u^2 + (v^2 - 2uv + u^2)$, so (iv) $\boldsymbol{v^2 = u^2 + 2as}$

$$s = \frac{1}{2}(u + v)t$$

$$s = ut + \frac{1}{2}at^2$$

$$v = u + at$$

$$v^2 = u^2 + 2as$$

Equations of motion with uniform acceleration

Sign convention

Quantity	To the left	To the right
Displacement, s	−ve	+ve
Velocity, v	−ve	+ve

- Acceleration: +ve; deceleration: −ve.
- Special case, uniform velocity: $u = v \Rightarrow a = 0$ and $s = ut$.

Example 2.2.1

An object is moving along the x-axis. Its velocity at the starting point O is 5 m s^{-1}. If it has a uniform acceleration of 6 m s^{-2}, calculate (a) its position after 2 s, (b) its velocity after 3 s.

Solution

Given: $u = 5$ m s^{-1}, $a = 6$ m s^{-2}

(a) $s = ut + \frac{1}{2}at^2 = (5)(2) + \frac{1}{2}(6)(2)^2 =$ **22 m** (to the right)

(b) $v = u + at = 5 + (6)(3) =$ **23 m s^{-1}** (to the right)

Example 2.2.2

An airplane lands with an initial velocity of 70.0 m/s and then decelerates at 1.50 m/s² for 40.0 s. What is its final velocity?

Solution

Given: $u = 70.0$ m/s, $a = -1.50$ m/s², $t = 40.0$ s

$v = u + at = 70.0 + (-1.50)(40.0) =$ **10.0 m/s**

Example 2.2.3

A vehicle accelerates uniformly along a straight road, increasing its speed from 4.0 m/s to 30.0 m/s in 13 s. Calculate (a) its acceleration, (b) the distance it moves in this time.

Solution

(a) $a = \frac{v - u}{t} = \frac{30.0 - 4.0}{13} =$ **2.0 m/s²**

(b) $s = \frac{1}{2}(u + v)t = \frac{1}{2}(4.0 + 30.0)(13) = 221 \approx$ **220 m**

Example 2.2.4

A body with initial velocity 8.0 m/s moves along a straight line with constant acceleration and travels 640 m in 40 s. For the 40 s interval, find (a) the average velocity, (b) the final velocity, (c) the acceleration.

Solution

Given: u = 8.0 m/s, s = 640 m, t = 40 s

(a) $\langle \vec{v} \rangle = \frac{s}{t} = \frac{640}{40} =$ **16 m/s**

(b) $\langle \vec{v} \rangle = \frac{u + v}{2} \Rightarrow v = 2(16) - 8.0 =$ **24 m/s**

(c) $a = \frac{v - u}{t} = \frac{24 - 8.0}{40} =$ **0.40 m/s²**

Exercises: equations of motion

1. A vehicle on a straight downhill road accelerated uniformly from 4.0 m/s to 29 m/s over a distance of 850 m, when the driver braked and stopped the vehicle in 28 s.
   (a) Calculate (i) the time taken to reach 29 m/s from 4 m/s, (ii) its acceleration during this time.
   (b) Calculate (i) the distance it travelled during deceleration, (ii) its average deceleration as it slowed down.

   Answer

   (a)(i) 52 s (ii) 0.49 m/s² (b)(i) 406 m (ii) −1.04 m/s²

   FULL SOLUTION

   (a)(i) u = 4.0 m s^{-1}, v = 29 m s^{-1}, s = 850 m

   s = ½(u + v)t ⇒ 850 = ½(4.0 + 29)t

   t = 850 / 16.5 = **51.5 s ≈ 52 s**

   (a)(ii) a = (v − u)/t = (29 − 4.0)/51.5

   = **0.49 m s^{-2}**

   (b)(i) u = 29 m s^{-1}, v = 0, t = 28 s

   s = ½(u + v)t = ½(29 + 0)(28)

   = **406 m**

   (b)(ii) a = (v − u)/t = (0 − 29)/28

   = **−1.04 m s^{-2}** (deceleration of 1.04 m s^{-2})

2. A rail wagon moving at 2.0 m/s on a level track reached a steady incline, which slowed it down in 15.0 s and caused it to reverse. Calculate (a) the distance it moved up the incline, (b) its acceleration on the incline, (c) its velocity and position on the incline after 20.0 s.

Answer

(a) 15 m (b) −0.133 m/s² (c) 0.67 m/s downwards, 13.3 m from the start

FULL SOLUTION

u = 2.0 m s^{-1} up the incline; it stops (v = 0) after t = 15.0 s.

(a) s = ½(u + v)t = ½(2.0 + 0)(15.0)

= **15 m**

(b) a = (v − u)/t = (0 − 2.0)/15.0

= **−0.133 m s^{-2}** (directed down the incline)

(c) t = 20.0 s (taking up the incline as positive):

v = u + at = 2.0 + (−0.1333)(20.0) = −0.67 m s^{-1}

s = ut + ½at^2 = (2.0)(20.0) + ½(−0.1333)$(20.0)^2$ = 40.0 − 26.7 = 13.3 m

Velocity **0.67 m s^{-1} down the incline**; position **13.3 m up the incline from the start**.

2.1 Linear motion: free fall

Free fall

Free-fall motion is the motion of an object under the influence of the ***Earth's gravitational attraction*** only.

- An object in free fall has only gravity acting on it (***no air resistance or friction***).
- On Earth all free-falling objects have the same ***acceleration due to gravity, g*** = 9.81 m s^{-2}.
- The direction of g is **downward**.

$$s = ut - \frac{1}{2}gt^2$$

$$v = u - gt$$

$$v^2 = u^2 - 2gs$$

Upward is positive, so $a = -g$. Put g = 9.81 m s^{-2} into these equations: the minus sign is already in them.

Sign convention

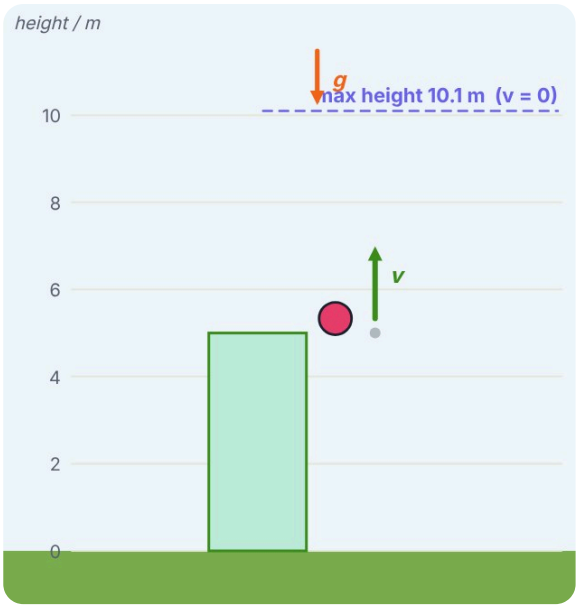
Quantity	Upward motion	Downward motion
Displacement, s	+ve	−ve
Velocity, v	+ve	−ve
Acceleration due to gravity	−ve	−ve

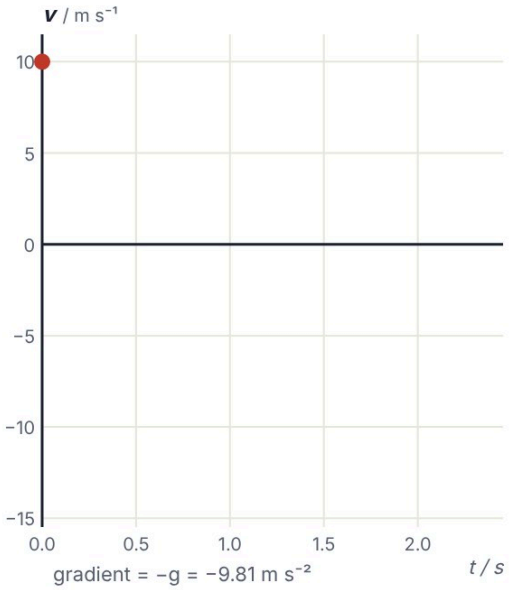
An object released from rest ($u = 0$) at height h has displacement $-s$ when it reaches the ground.

Simulation

Vertical motion: drop it, throw it up, launch it from a height

Set the launch velocity (upward is positive) and the starting height, or tap one of the examples. The ball, its velocity arrow and the v–t graph run together; the results appear when it lands.





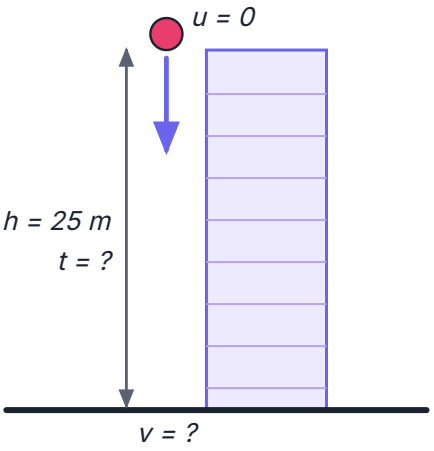
t	displacement s	velocity v	max height (above ground)	time to top
0.00 s	**0.00 m**	**10.00 m/s**	**10.10 m**	**1.02 s**

time of flight	impact velocity
…	…

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 2 to use the simulation.

Example 2.3.1

An object is released from rest at a height of 25 m from a building. Determine the time and velocity when it reaches the ground.



Solution

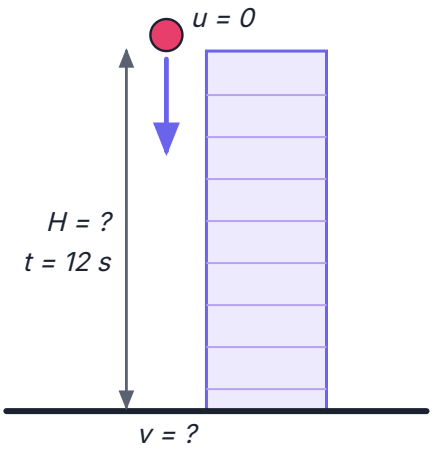
$u = 0$, $s = -25$ m, $g = 9.81\ \text{m s}^{-2}$

$s = ut - \frac{1}{2}gt^2$: $-25 = 0 - \frac{1}{2}(9.81)t^2$, $t = \mathbf{2.26\ s}$

$v = u - gt = 0 - (9.81)(2.26) = \mathbf{-22.2\ m s^{-1}}$ (−ve: downward)

Example 2.3.2

An object is released from rest at height H from a building; it takes 12 s to reach the ground. Determine the height and the velocity when it reaches the ground.



Solution

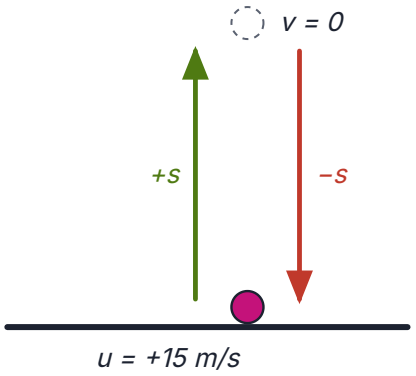
$u = 0$, $t = 12$ s, $g = 9.81\ \text{m s}^{-2}$

$s = ut - \frac{1}{2}gt^2 = 0 - \frac{1}{2}(9.81)(12)^2 = -706$ m, so the height is **706 m**

$v = u - gt = 0 - (9.81)(12) =$ **$-118\ \text{m s}^{-1}$** (−ve: downward)

Example 2.3.3

An object is thrown upward with a velocity of 15 m s^{-1}. What is its **final velocity** when it reaches the ground? What is the **flight time**?



Solution

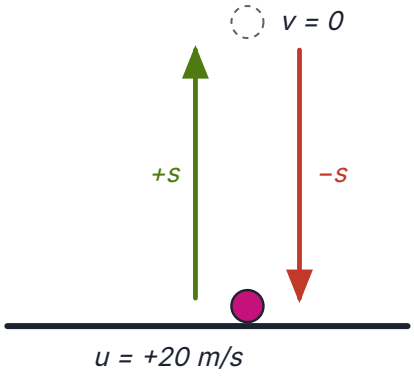
Back on the ground: displacement $s = 0$; $u = +15\ \text{m s}^{-1}$

$v^2 = u^2 - 2gs = (15)^2 - 2(9.81)(0) \Rightarrow v = \mathbf{-15\ m s^{-1}}$ (moving down)

$s = ut - \frac{1}{2}gt^2$: $0 = 15t - \frac{1}{2}(9.81)t^2 \Rightarrow t = \mathbf{3.06\ s}$

Example 2.3.4

A ball is thrown vertically upward with $u = 20\ \text{m s}^{-1}$. Determine the **maximum height** and the **time taken to reach the maximum height**.



Solution

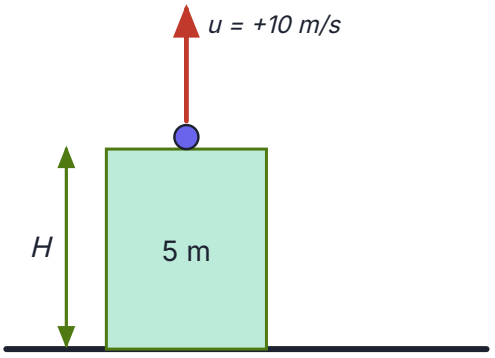
At maximum height, $v = 0$. $u = 20\ \text{m s}^{-1}$, $g = 9.81\ \text{m s}^{-2}$

$v^2 = u^2 - 2gs$: $0 = 20^2 - 2(9.81)s \Rightarrow h =$ **20.4 m**

$v = u - gt$: $0 = 20 - 9.81t \Rightarrow t =$ **2.04 s**

Example 2.3.5

An object is launched upward from a table of height 5 m with velocity 10 m s^{-1}. Determine (a) the **maximum height**, (b) the **time of flight**, (c) its **velocity** when it hits the ground.



Solution

(a) At maximum height $v = 0$: $0 = 10^2 - 2(9.81)s \Rightarrow s = 5.10$ m above the table.

Maximum height above the ground, $h = s + H = 5.10 + 5 =$ **10.10 m**

(b) Ground is 5 m below the launch point: $s = -5$ m.

$-5 = 10T - \frac{1}{2}(9.81)T^2 \Rightarrow T =$ **2.45 s**

(c) $v = u - gt = 10 - (9.81)(2.45) = -14.03$ m s^{-1}

or $v^2 = u^2 - 2gs = 10^2 - 2(9.81)(-5) \Rightarrow v = \pm 14.07$, take **−14.07 m s^{-1}** (moving downward)

The two answers differ slightly only because T was rounded to 2.45 s. The v^2 method does not need T, so it is the more accurate one.

Exercises: free fall

1. A pebble, released at rest from a bridge, took 0.9 s to hit the water. Calculate (a) the distance it fell before hitting the water, (b) its speed just before hitting the water.

Answer

(a) 4.0 m (b) 8.8 m/s

FULL SOLUTION

u = 0, t = 0.9 s, a = g = 9.81 m s^{-2} (downward positive)

(a) s = ut + ½gt^2 = 0 + ½(9.81)(0.9)^2

= 3.97 m ≈ **4.0 m**

(b) v = u + gt = 0 + (9.81)(0.9)

= **8.8 m s^{-1}**

2. A spanner was dropped from a hot-air balloon when the balloon was at rest 50 m above the ground. Calculate (a) the time taken for the spanner to hit the ground, (b) the speed of impact of the spanner on hitting the ground.

Answer

(a) 3.2 s (b) 31 m/s

FULL SOLUTION

The balloon is at rest, so u = 0; s = 50 m (downward positive).

(a) s = ½gt^2 ⇒ t = √(2s/g) = √(2 × 50 / 9.81)

= **3.2 s**

(b) v^2 = u^2 + 2gs = 0 + 2(9.81)(50) = 981

v = **31 m s^{-1}**

3. A stone is thrown vertically upward with a speed of 12.0 m/s from the edge of a cliff 75.0 m high. (a) How much later does it reach the bottom of the cliff? (b) What is its speed just before hitting? (c) What total distance did it travel?

Answer

(a) 5.33 s (b) 40.2 m/s (c) 89.7 m

FULL SOLUTION

Take upward as positive: u = +12.0 m s^{-1}, a = −9.81 m s^{-2}, and the bottom of the cliff is at s = −75.0 m.

(a) s = ut + ½at^2: −75.0 = 12.0t − 4.905t^2

4.905t^2 − 12.0t − 75.0 = 0

t = [12.0 + √(12.0^2 + 4(4.905)(75.0))] / (2 × 4.905) = (12.0 + 40.19)/9.81

= **5.32 s**

Note: the notes give 5.33 s; the difference is only rounding.

(b) v^2 = u^2 + 2as = 12.0^2 + 2(−9.81)(−75.0) = 1615.5

|v| = **40.2 m s^{-1}**

(c) Height risen above the cliff top: v^2 = u^2 + 2as ⇒ 0 = 12.0^2 − 2(9.81)h

h = 144 / 19.62 = 7.34 m

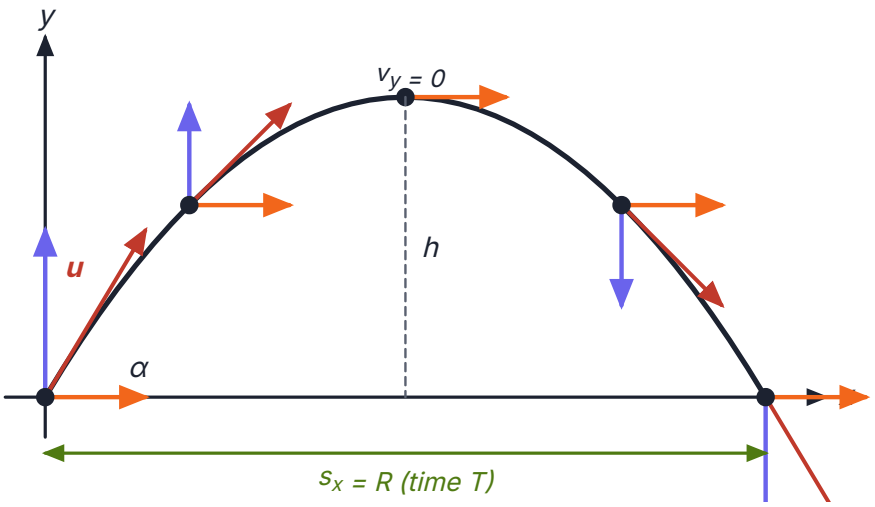
Total distance = 7.34 (up) + 7.34 (back down) + 75.0 = **89.7 m**

2.2 Projectile motion

Projectile motion

is a combination of ***horizontal motion with constant velocity*** and ***vertical motion with constant acceleration***.

- **Horizontal**: the velocity in the **x-direction is constant** (v_x = constant, $a_x = 0$).
- **Vertical**: the velocity in the **y-direction has a constant acceleration due to gravity**, air resistance ignored ($a_y = -g = -9.81\ \mathrm{m\,s^{-2}}$).

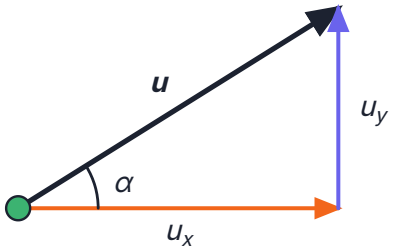


u_x horizontal component (same everywhere) — v_y vertical component — velocity v (along the tangent)

Parabola: the path is symmetrical.

u = initial velocity; v = velocity at time t; u_x, u_y = initial x- and y-components; v_y = y-component at time t; s_x = R = horizontal range; α (or θ) = launch angle; h = maximum height; T = time of flight.

Equations of projectile motion



$$u_x = u\cos\alpha$$

$$u_y = u\sin\alpha$$

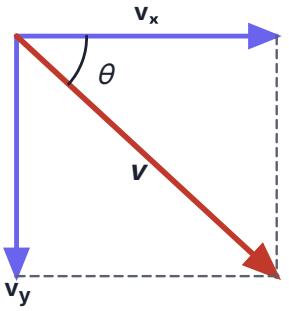
$$\alpha = \tan^{-1}\frac{u_y}{u_x}$$

Quantity	Horizontal (x-component)	Vertical (y-component)
Displacement, s	$s_x = u_x t$	$s_y = u_y t - \frac{1}{2} g t^2$
Velocity, v	$v_x = u_x$ (constant)	$v_y = u_y - gt$ $v_y^2 = u_y^2 - 2gs_y$
Acceleration, a	$a_x = 0$	$a_y = -g = -9.81\ \text{m s}^{-2}$ (constant)

Velocity at any point

Magnitude: $|v| = \sqrt{v_x^2 + v_y^2}$

Direction: $\theta = \tan^{-1} \frac{v_y}{v_x}$



Maximum height, H

Vertically: $u_y = u \sin\theta$. At maximum height $v_y = 0$.

$$v_y^2 = u_y^2 - 2gs_y \Rightarrow 0 = (u \sin\theta)^2 - 2gH$$

$$H = \frac{u^2 \sin^2\theta}{2g}$$

Time of flight, T

Vertically: at the highest point $v_y = 0$. Using $v_y = u_y - gt$: $0 = u \sin\theta - gt$, so the time to reach the maximum height is

$$t = \frac{u \sin\theta}{g}$$

The path is symmetrical: **time to go up = time to come down**.

$$T = 2t = \frac{2u \sin \theta}{g}$$

Horizontal range, R

Horizontally: $a_x = 0$, $u_x = u \cos \theta$, $s_x = R = u_x T$

$$R = (u \cos \theta)\left(\frac{2u \sin \theta}{g}\right) \quad \text{where } \sin 2\theta = 2 \sin \theta \cos \theta$$

$$R = \frac{u^2 \sin 2\theta}{g}$$

The horizontal range is the horizontal displacement of the projectile (back to the launch level).

Maximum range and maximum height

$$H = \frac{u^2 \sin^2 \theta}{2g}$$

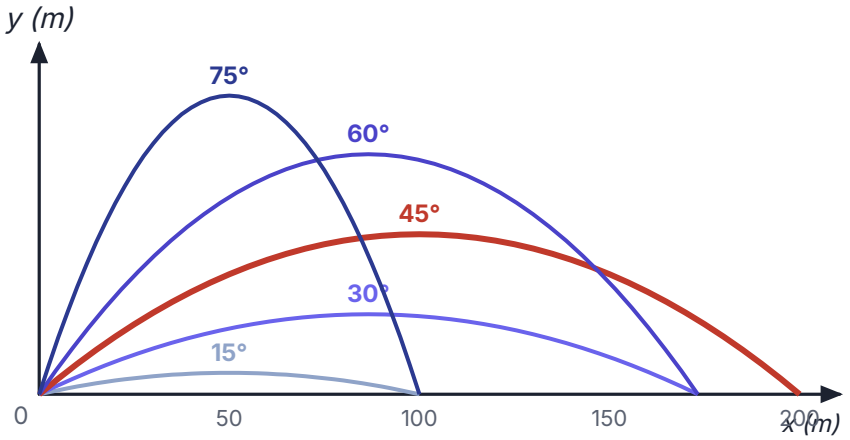
maximum when **θ = 90°**:

$$H_{max} = \frac{u^2}{2g}$$

$$R = \frac{u^2 \sin 2\theta}{g}$$

maximum when 2θ = 90°, i.e. **θ = 45°**:

$$R_{max} = \frac{u^2}{g}$$

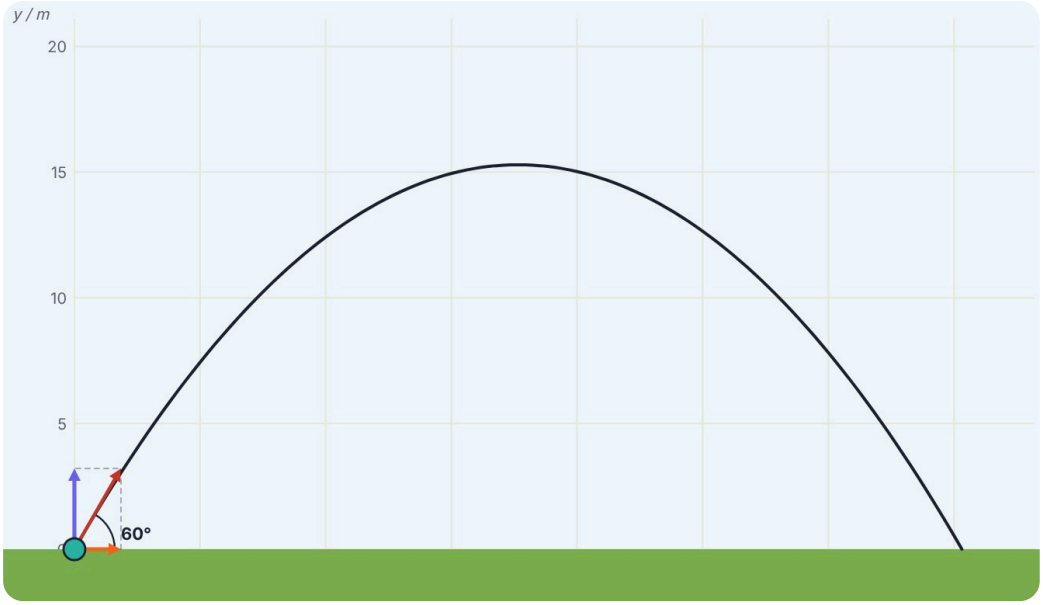


Same launch speed, different angles: 45° goes furthest; 30° and 60° (and 15° and 75°) land at the same place.

Simulation

Projectile launcher

Change the launch speed, angle and height. Watch the horizontal component stay the same while the vertical component shrinks, reverses and grows. "Compare angles" draws 15° to 75° together; "Air resistance" shows the real, lopsided path.



t	v_x (horizontal)	v_y (vertical)	speed $\lvert v \rvert$	max height H
0.00 s	10.00 m/s	17.32 m/s	20.00 m/s	15.29 m

time of flight T	range R	impact
3.53 s	35.31 m	20.0 m/s, 60.0° below

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 2 to use the simulation.

Example 2.4.1

A ball is launched with a velocity of 20 m s^{-1} at an angle of 60° with the horizontal at point A. Calculate (a) the **maximum height**, H, (b) the **time of flight**, (c) the **horizontal range**, R, (d) its **velocity** when it strikes the ground.

Solution

(a) Vertically: $u_y = 20 \sin 60°$, $v_y = 0$ at maximum height.

$v_y^2 = u_y^2 - 2gs_y$: $0 = (20 \sin 60°)^2 - 2(9.81)H \Rightarrow H =$ **15.29 m**

(b) Vertically: $s_y = 0$ (back on the ground). $0 = u \sin\theta\, t - \frac{1}{2}gt^2 \Rightarrow T = \frac{2u \sin\theta}{g} = \frac{2(20 \sin 60°)}{9.81}$

= **3.53 s**

(c) Horizontally: $u_x = 20 \cos 60°$, $a_x = 0$. $R = u_x T = (20 \cos 60°)(3.53) =$ **35.3 m**

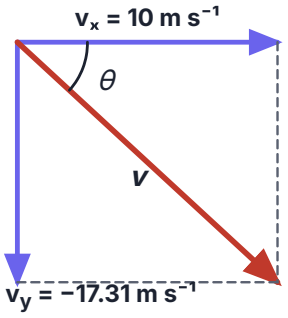
(d) $v_x = u_x = 20 \cos 60° = 10$ m s^{-1}

$v_y = u_y - gT = 20 \sin 60° - (9.81)(3.53) = -17.31$ m s^{-1} (downward)

Magnitude: $v = \sqrt{10^2 + (-17.31)^2}$ = **20 m s^{-1}**

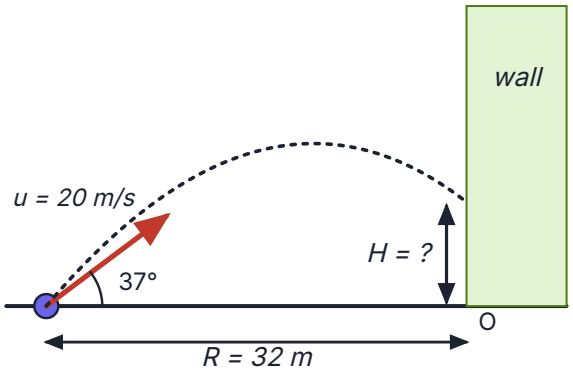
Direction: $\theta = \tan^{-1}\frac{-17.31}{10} = -60°$

Its velocity is 20 m/s at 60° below the x-axis.



Example 2.4.2

Refer to the figure. Where will the ball hit the wall?



Solution

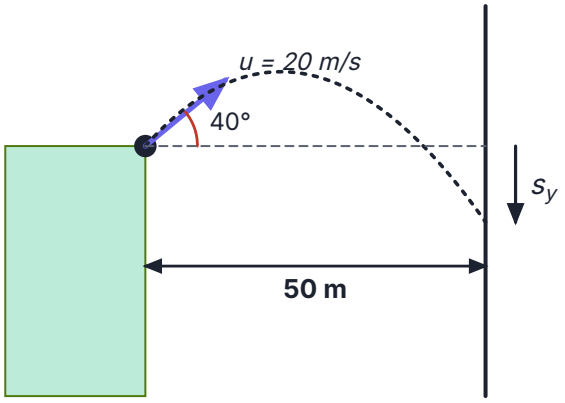
Horizontally: $u_x = 20 \cos 37°$, $s_x = 32$ m. $s_x = u_x t$: $32 = (20 \cos 37°)t \Rightarrow t = 2.0$ s

Vertically: $u_y = 20 \sin 37°$. $s_y = u_y t - \frac{1}{2}gt^2 = (20 \sin 37°)(2.0) - \frac{1}{2}(9.81)(2.0)^2 = 4.5$ m

The ball hits the wall **4.5 m above point O**.

Example 2.4.3

A ball is thrown from the top of one building toward a tall building 50 m away. The initial velocity of the ball is 20 m/s at 40° above the horizontal. How far above or below its original level will the ball strike the opposite wall?



Solution

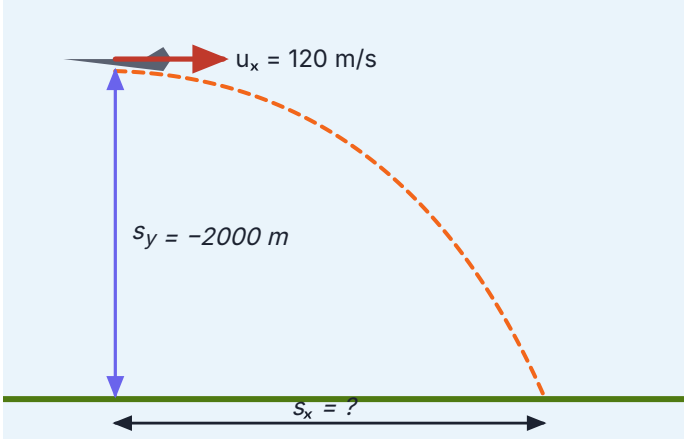
Horizontally: $u_x = 20 \cos 40°$, $s_x = 50$ m. $50 = (20 \cos 40°)t \Rightarrow t = 3.26$ s

Vertically: $u_y = 20 \sin 40°$. $s_y = (20 \sin 40°)(3.26) - \frac{1}{2}(9.81)(3.26)^2 = -10.22$ m

The ball hits **10.22 m below** its original level.

Example 2.4.4

A military airplane is flying horizontally at 120 m/s and accidentally drops a bomb at an elevation of 2000 m. (a) How much time is required for the bomb to reach the earth? (b) How far does it travel horizontally while falling? (c) Find the horizontal and vertical components of its velocity just before it strikes the earth. (d) Find the resultant velocity of the bomb just before it strikes the earth.



Solution

(a) Vertically: $u_y = 120 \sin 0° = 0$, $s_y = -2000$ m.

$-2000 = 0 - \frac{1}{2}(9.81)t^2 \Rightarrow t =$ **20.2 s**

(b) Horizontally: $u_x = 120 \cos 0° = 120$ m/s. $s_x = u_x t = (120)(20.2) =$ **2424 m**

(c) $v_x = 120\ \text{m s}^{-1}$; $v_y = u_y - gt = 0 - (9.81)(20.2) =$ **−198 m s^{-1}** (downward)

(d) Magnitude: $v = \sqrt{120^2 + (-198)^2} =$ **231.5 m s^{-1}**

Direction: $\theta = \tan^{-1}\frac{-198}{120} = -58.8°$

The bomb hits the ground at 231.5 m/s, 58.8° below the x-axis.

Exercises: projectile motion

1. A body is projected upward from level ground at an angle of 50° with the horizontal with an initial speed of 40 m/s. (a) How long will it take to hit the ground? (b) How far from the starting point will it strike? (c) At what angle with the horizontal will it strike?

Answer

(a) 6.3 s (b) 0.16 km (c) 50°

FULL SOLUTION

$u_x = 40 \cos 50° = 25.71 \text{ m s}^{-1}$, $u_y = 40 \sin 50° = 30.64 \text{ m s}^{-1}$

(a) Lands at the same level ($s_y = 0$): $0 = u_y t - \frac{1}{2}gt^2$

$t = 2u_y/g = 2(30.64)/9.81 = 6.25 \text{ s} \approx$ **6.3 s**

(b) $R = u_x t = 25.71 \times 6.25$

$= 161 \text{ m} \approx$ **0.16 km**

(c) On landing $v_y = -30.64 \text{ m s}^{-1}$ and $v_x = 25.71 \text{ m s}^{-1}$ (by symmetry)

$\tan \theta = 30.64 / 25.71 \Rightarrow \theta =$ **50° below the horizontal**

2. An object is projected horizontally at 16 m/s into the sea from a cliff top 45.0 m high. Calculate (a) how long it takes to reach the sea, (b) the horizontal range, (c) its impact velocity.

Answer

(a) 3.0 s (b) 49 m (c) 34 m/s at 62° to the horizontal

FULL SOLUTION

$u_x = 16 \text{ m s}^{-1}$, $u_y = 0$, vertical drop = 45.0 m

(a) $s_y = \frac{1}{2}gt^2 \Rightarrow t = \sqrt{(2 \times 45.0 / 9.81)}$

$=$ **3.0 s** (3.03 s)

(b) $R = u_x t = 16 \times 3.03$

$=$ **49 m**

(c) $v_y = gt = 9.81 \times 3.03 = 29.7 \text{ m s}^{-1}$, $v_x = 16 \text{ m s}^{-1}$

$v = \sqrt{(16^2 + 29.7^2)}$

$=$ **34 m s^{-1}**

$\tan \theta = 29.7 / 16 \Rightarrow \theta =$ **62° below the horizontal**

3. An object is thrown at 37° with the horizontal from a bridge 20.0 m above a river, with initial speed 30.0 m/s. Calculate (a) the time of flight, (b) the horizontal range, (c) the final velocity at impact.

Answer

(a) 4.57 s (b) 110 m (c) 35.9 m/s, 48.1° below the horizontal

FULL SOLUTION

$u_x = 30.0 \cos 37° = 23.96 \text{ m s}^{-1}$, $u_y = 30.0 \sin 37° = 18.05 \text{ m s}^{-1}$ (upward positive)

(a) The river is 20.0 m below: $-20.0 = 18.05t - 4.905t^2$

$4.905t^2 - 18.05t - 20.0 = 0$

$t = [18.05 + \sqrt{(18.05^2 + 4(4.905)(20.0))}] / 9.81 = (18.05 + 26.80)/9.81$

$=$ **4.57 s**

(b) $R = u_x t = 23.96 \times 4.57$

$=$ **110 m** (109.5 m)

(c) $v_y = u_y - gt = 18.05 - 9.81(4.572) = -26.80 \text{ m s}^{-1}$

$v = \sqrt{(23.96^2 + 26.80^2)}$

$=$ **35.9 m s^{-1}**

$\tan\theta = 26.80 / 23.96 \Rightarrow \theta =$ **48.2° below the horizontal**

Note: the notes give 48.1°; the difference is only rounding.

4. An arrow is fired at 45 m/s at 30° above the horizontal. Calculate (a) its maximum height, (b) how long it takes to reach the maximum height, (c) how long it takes to return to the same horizontal level as it started, (d) the horizontal distance travelled to the point of return in (c).

Answer

(a) 25.8 m (b) 2.3 s (c) 4.6 s (d) 179 m

FULL SOLUTION

$u_x = 45\cos 30° = 38.97\ \text{m s}^{-1}$, $u_y = 45\sin 30° = 22.5\ \text{m s}^{-1}$

(a) At maximum height $v_y = 0$: $v_y^2 = u_y^2 - 2gH$

$H = u_y^2/(2g) = 22.5^2 / (2 \times 9.81)$

$=$ **25.8 m**

(b) $v_y = u_y - gt = 0 \Rightarrow t = 22.5 / 9.81$

$=$ **2.3 s** (2.29 s)

(c) By symmetry, time to return to the same level $= 2 \times 2.29$

$=$ **4.6 s** (4.59 s)

(d) $R = u_x t = 38.97 \times 4.59$

$=$ **179 m**

5. A ball is thrown upward at 30° to the horizontal and lands on the top edge of a building 20 m away. The top edge is 5.0 m above the throwing point. How fast was the ball thrown?

Answer

20 m/s

FULL SOLUTION

Let the speed of projection be u. $u_x = u\cos 30°$, $u_y = u\sin 30°$.

Horizontal: $20 = (u\cos 30°)t \Rightarrow t = 20/(0.866u)$

Vertical: $5.0 = (u\sin 30°)t - \tfrac{1}{2}gt^2$

Substitute $(u\sin 30°)t = 20\tan 30° = 11.55$ m:

$5.0 = 11.55 - 4.905t^2 \Rightarrow t^2 = 1.335 \Rightarrow t = 1.155$ s

$u = 20 / (0.866 \times 1.155)$

$=$ **20 m s^{-1}**

2.2 Air resistance effects on projectile motion

Without air resistance

- The trajectory is a perfect parabola.
- Horizontal velocity is constant; vertical acceleration = g.
- Maximum range is achieved at launch angle $\theta = 45°$.

With air resistance

- **Horizontal velocity decreases**: air drag opposes the horizontal motion.
- **Vertical deceleration increases** on the way up: net downward force = mg + drag.
- **The trajectory is no longer a symmetrical parabola.**
- **Maximum range occurs at an angle less than 45°.**
- **Horizontal range and maximum height are both reduced.**
- The time to descend is longer than the time to ascend.
- The impact speed is less than the launch speed.

Teacher's note

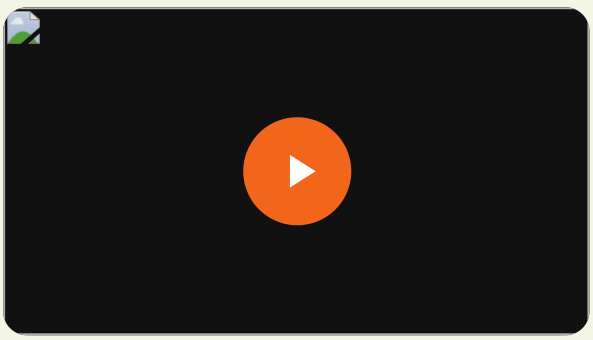
Open the projectile launcher above and switch on "Air resistance": the dashed path is the ideal parabola, the solid path is with drag. Try 45° and then 40°: with drag, 40° goes further.

PHYSICS IN REAL LIFE

STPM 960 · Semester 1 · 2.2 Projectiles

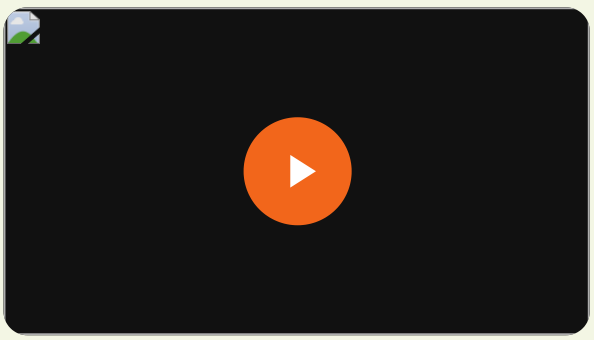
Projectile Motion in Sports

Once the ball or the athlete leaves the ground or the hand, it becomes a projectile. If air resistance is neglected, the **horizontal component of velocity is constant** and the **vertical acceleration is g downwards**.



Long jump Projectile Motion Problem | Physics of Long Jump!

Open in YouTube ↗



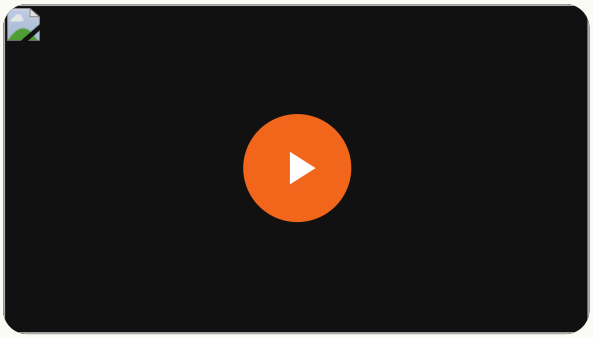
3.3 Projectile Motion, Part B Open in YouTube ↗

Football kick Main lesson

Long jump Main lesson

After take-off, treat the athlete's centre of mass as a projectile: launch velocity, time in the air and horizontal range.

A worked example with a kicked football: maximum height, time of flight and range. Watch it after learning the equations.

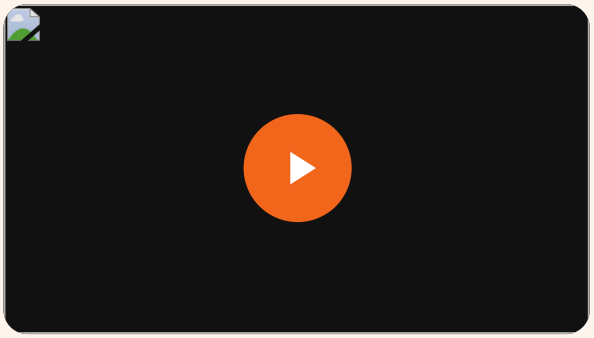


Projectile Motion - Basketball in Projectile Motion

Open in YouTube ↗

Basketball shot

The ball follows a curved path once it leaves the player's hands; compare its horizontal and vertical motion.



Physics of Javelin Throw is more than just Projectile Motion | Neeraj Chopra Olympic Gold & Physics

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Javelin throw Real-world extension

Release speed, angle and height all matter, but air resistance and aerodynamic lift act on a real javelin, so its path is not a perfect parabola.

While you watch, identify:

1. the **launch point**: the instant the object becomes a projectile;
2. the **horizontal motion**: which velocity component stays constant;
3. the **vertical acceleration**: its size and direction;
4. **one factor** that makes the real path differ from the ideal model.

Real sports trajectories are not always perfect parabolas. A spinning football can curve sideways because of the Magnus effect, and a javelin is affected by lift and air resistance; these go beyond the basic projectile model.